Person Behaviour Analytics Platform

Production Architecture Overview

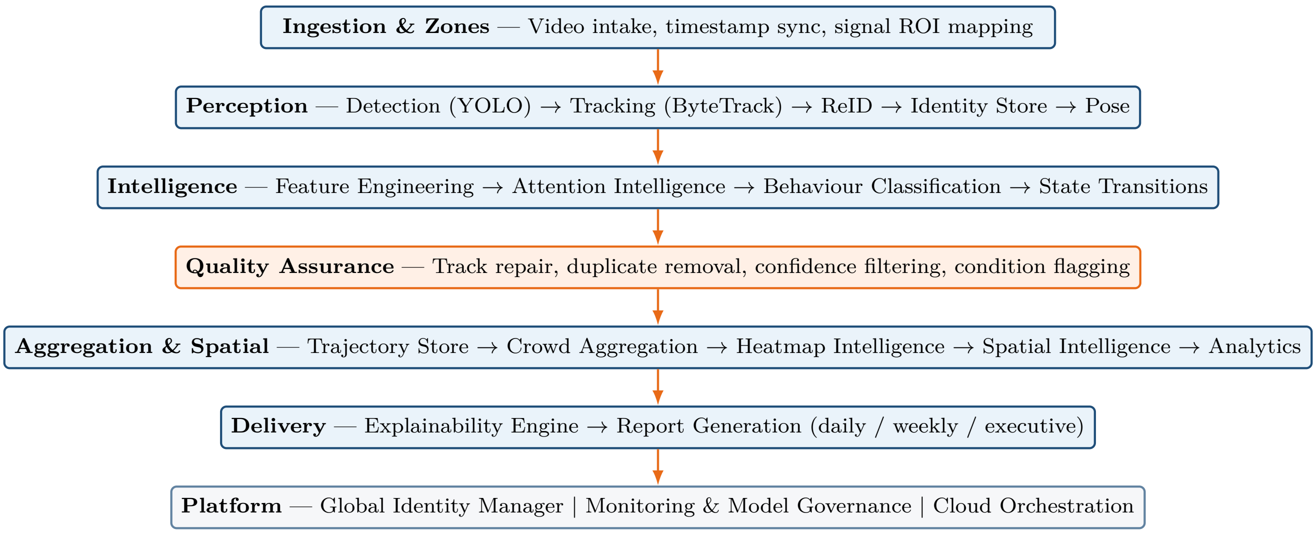
Traffic-Signal CCTV Intelligence | Client & Engineering Reference

Executive Summary

This document describes the production architecture for pedestrian behaviour analytics at traffic-signal and adjacent retail CCTV sites. The platform ingests weekly HDD uploads from multiple cameras, resolves persistent pedestrian identities, classifies behavioural states, aggregates crowd intelligence, generates spatial heatmaps, and delivers explainable commercial reports comparable to Tactical Hive, BriefCam, RetailNext, and Irisity.

The legacy pipeline (Ingestion → Detection → Tracking → Pose → Classification → Analytics → PDF) is replaced by a layered distributed system with ReID-based identity management, engineered feature inputs, attention and queue intelligence, mandatory quality assurance, crowd and spatial analytics, explainability, monitoring, and cloud orchestration. The design prioritises accuracy, scalability, explainability, and commercial viability.

High-Level Architecture



Design principles: (1) classifiers consume engineered features, not raw pose; (2) no metric reaches reports without QA approval; (3) identity is resolved before dwell, funnel, and transition analytics; (4) every report metric is traceable via the Explainability Engine.

Core Modules

Each module below is a first-class production component. Descriptions are intentionally concise; implementation detail lives in engineering runbooks.

Video Ingestion & Zone Mapping

Accepts HDD/RTSP uploads, validates integrity, synchronises timestamps, and maps signal zones (Approach, Threshold, Queue Strip, Dwell Bay, Exit, Attention ROI). Zone context drives all downstream spatial and behavioural logic.

Person Detection

YOLOv11 detects pedestrians within the signal footprint. Low-confidence and undersized detections are filtered before tracking.

Short-Term Tracking

ByteTrack/BoT-SORT assigns local track IDs and produces tracklets with trajectories. Tracklets are short-lived associations—not final person identities.

Person ReID
Extracts appearance embeddings from quality tracklet frames (FastReID/OSNet). Matches against a gallery to recover identity after occlusion, vehicle blocking, crowd hiding, or ROI exit/re-entry. Reduces duplicate visitor counts and stabilises dwell, waiting-time, and funnel metrics.

Persistent Identity Store
Maintains canonical person identities with confidence scoring. Merges fragmented tracklets, prevents duplicate counting, and feeds all person-level analytics. Supports per-camera and site-wide identity scopes.

Pose Extraction
Extracts skeleton keypoints for head/torso orientation and motion primitives. Pose feeds the Feature Engineering Layer only—never the behaviour classifier directly.

Feature Engineering
Transforms pose, trajectory, and zone context into structured features: speed, acceleration, dwell duration, heading, turn frequency, trajectory smoothness, zone occupancy, attention score, queue score, neighbour/crowd density, stop duration, and movement entropy.

Attention Intelligence
Replaces naive head-direction "Looking" detection. Combines head orientation, body orientation, trajectory slowdown, dwell time, movement hesitation, directional consistency, and spatial context to estimate pedestrian attention on a 0–100 scale. Outputs tiers: High, Medium, Low, and Unclear attention.

Queue Intelligence
Scores queue likelihood using spacing consistency, directional alignment, waiting duration, movement synchronisation, queue persistence, and density—not spacing alone. Rejects social groups, random gatherings, and unstructured crowds.

Behaviour Classification
Temporal model (TCN/Transformer) consumes engineered features and zone context. Classifies Walking, Standing, Attention, Waiting, Queueing, Entering, Exiting, and Unclear states with confidence gating.

State Transition Engine
Models behaviour sequences (e.g. Walking → Attention → Waiting → Queueing → Entering → Exiting). Produces transition matrices, conversion funnels, abandonment rates, and dwell funnels for business intelligence.

Quality Assurance Engine
Mandatory gate before analytics. Handles fragmented track repair, duplicate removal, identity conflict resolution, low-confidence filtering, and condition flagging (rain, glare, night, occlusion, camera shake). Outputs a report-level Quality Score included in every deliverable.

Trajectory Store
Persists QA-approved movement paths, zone crossings, dwell segments, and state timelines. Single source for heatmap and spatial intelligence generation.

Crowd Aggregation Engine
Transforms person-level events into crowd metrics: behaviour mix, hourly/day/weekly distributions, behaviour lift, dwell distributions, crowd and queue density, and attention distributions. Uses person-seconds for mix calculations and deduplicated identities for footfall.

Heatmap Intelligence Engine
Generates pedestrian density, attention, dwell, trajectory, congestion, dead-zone, and hotspot maps via KDE, spatial density analysis, and trajectory clustering. Outputs stored as report artifacts linked to analytics.

Spatial Intelligence Layer
Classifies attention zones, dead zones, congestion zones, crossing zones, high-dwell regions, and low-engagement regions. Delivers spatial opportunity insights, signage placement recommendations, and traffic-flow analysis.

Analytics Engine
Computes BI metrics: trends, lift, funnels, commercial opportunity signals, and executive narrative summaries from crowd and spatial aggregates.

Explainability Engine
Traces every report metric to supporting evidence: confidence scores, behavioural reasoning, track samples, and quality indicators. Enables audit, client trust, and regulatory review.

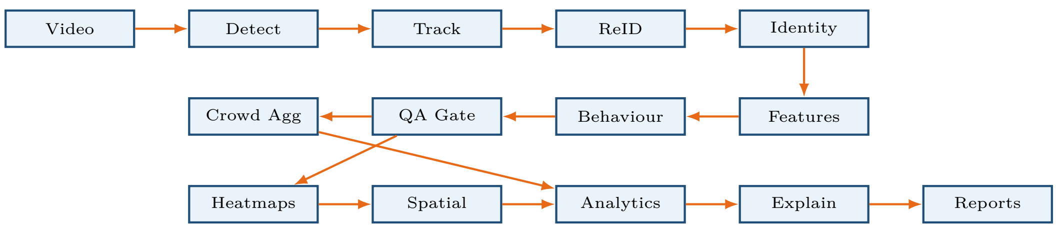
Report Generation
Renders Behaviour Analytics, Spatial Intelligence, Heatmaps, Crowd Insights, Attention Analysis, Queue Analytics, Conversion Funnels, Commercial Opportunity, and Weekly Executive Dashboards—daily and weekly, per camera and site-wide.

Global Identity Manager
Reconciles identities across Camera A/B/C using cross-camera ReID and zone-transition rules. Enables deduplicated site footfall and cross-camera journey analytics alongside per-camera reports.

Monitoring & Model Governance
Tracks model drift, data drift, confidence levels, tracking and behaviour accuracy proxies, processing health, and anomaly alerts. Provides model health and processing health dashboards.

Cloud Orchestration
FastAPI orchestrator, Redis job queues, RunPod GPU workers, PostgreSQL, object storage, and asynchronous report workers. Supports horizontal scaling, fault-tolerant retries, and weekly multi-camera batch processing.

Core Data Flow



Video flows through perception and intelligence stacks. QA-approved events feed crowd aggregation, heatmaps, and spatial intelligence in parallel. Analytics results pass through explainability before report generation. Global Identity Manager operates across camera pipelines at the platform layer.

Core Database Entities

The schema supports identity lineage, event auditability, and report traceability. Key entities:

- **identities** — Canonical person records; links fragmented tracklets; stores identity confidence and site-level mapping.
- **tracks** — Short-term tracklets with local track IDs, time spans, and quality scores.
- **trajectory_points** — QA-approved movement paths with zone membership and state timeline.
- **engineered_features** — Structured feature vectors per person per time window; classifier input audit trail.

- **behaviour_events** — Classified behavioural states with confidence and zone context.
- **transition_events** — State-to-state transitions for funnel and conversion analytics.
- **attention_events** — Attention scores and tiers (High/Medium/Low/Unclear).
- **queue_events** — Queue confidence scores and queue duration segments.
- **heatmaps** — Stored map artifacts (density, attention, dwell, trajectory, dead-zone, hotspot).
- **spatial_zones** — Camera zone definitions and versions (Approach, Queue, Threshold, etc.).
- **quality_metrics** — QA exclusion rates, condition flags, and report-level Quality Score.
- **daily_reports / weekly_reports** — Generated deliverables with linked explainability evidence.
- **camera_metadata** — Camera configuration, site membership, and processing status.

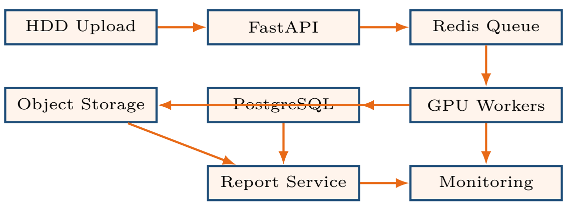
Business Capabilities

The platform delivers commercial-grade intelligence across the following report types:

- **Behaviour Analytics** — Behaviour mix, key state reads, behaviour lift, executive narrative.
- **Spatial Intelligence** — Zone classifications, dead zones, congestion, opportunity insights.
- **Heatmaps** — Density, attention, dwell, trajectory, and hotspot visualisations.
- **Crowd & Queue Analytics** — Footfall, density distributions, queue pressure, peak periods.
- **Attention Analysis** — High-attention zones linked to conversion performance.
- **Conversion Funnels** — Stage counts, drop-offs, abandonment, dwell funnels.
- **Commercial Opportunity** — High-attention / low-conversion zones; signage recommendations.
- **Weekly Executive Dashboard** — Site summary, trends, anomalies, quality score.

Reports are available per camera and site-wide. Each includes a Quality Score and explainability appendix.

Deployment Architecture



Weekly HDD uploads are copied to object storage. FastAPI creates per-camera processing jobs enqueued to Redis. RunPod GPU workers execute the perception and intelligence pipeline, writing events to PostgreSQL and artifacts to object storage. CPU report workers generate PDFs asynchronously. Monitoring tracks job health, model performance, and end-to-end SLA. The architecture scales horizontally by adding GPU and report workers; failed jobs retry with idempotent design and dead-letter handling. Global Identity Manager and site-level aggregation run after all camera jobs for a site-day complete.

Technology Stack

Layer	Technology
Orchestration	FastAPI, Redis Queue (Celery/RQ)
Detection	YOLOv11
Tracking	ByteTrack, BoT-SORT
ReID	FastReID (OSNet)
Pose	YOLO-Pose, RTMPose
Behaviour model	Temporal Transformer / TCN (PyTorch)
Heatmaps	SciPy KDE, OpenCV, DBSCAN clustering
Database	PostgreSQL
Object storage	S3-compatible (AWS S3, MinIO, RunPod volumes)
GPU compute	RunPod, AWS G5, Azure NC-series
Reporting	ReportLab, Matplotlib, Plotly
Monitoring	Prometheus, Grafana, MLflow

Key Risks & Mitigations

Risk	Impact	Mitigation
ReID false merge/split	Inaccurate footfall	**Identity confidence scoring, QA conflict resolution, conservative matching**
Attention false positives	Inflated ROI metrics	**Multi-signal Attention Intelligence; tier gating**
Queue misclassification	Wrong bottleneck signals	**Queue Intelligence; social-group rejection**
Low-quality footage	Unreliable labels	**QA condition flagging; segment exclusion; Quality Score on reports**
Model drift	Degrading accuracy over time	**Monitoring & Model Governance; retrain triggers**
Privacy exposure	Regulatory risk	**Pseudonymous IDs; no face storage by default**
Processing failures	Delayed deliverables	**Idempotent jobs, retries, dead-letter queue, horizontal workers**

Conclusion

This architecture delivers a production-ready pedestrian behaviour analytics platform for traffic-signal CCTV environments. All major capabilities—ReID identity management, engineered features, attention and queue intelligence, state-transition funnels, crowd and spatial analytics, heatmaps, explainability, multi-camera identity, quality assurance, monitoring, and distributed cloud processing—

are preserved in a modular, scalable design suitable for commercial deployment and client-facing delivery.